ChatGPT

Executive Summary

Ultra-wideband (UWB) and magneto-inductive (MI) systems offer complementary approaches for positioning drones in GPS-denied underground environments. UWB uses nanosecond-scale pulses (3.1–10.6 GHz) and Time-of-Flight (ToF) ranging to achieve centimeter-to-decimeter accuracy [1] [2]. Typical UWB modules (e.g. Decawave/Qorvo DWM1001) weigh ≈2–3 cm and provide ~10 cm ranging precision and ~60 m LOS range [2]. UWB tolerates multipath better than narrowband radios (pulses allow early-path detection [3]) but still requires line-of-sight and suffers from NLOS errors (laboratory tests report <0.3 m error in LOS vs ~2 m in NLOS [4]). In mines, complex tunnel geometry can degrade UWB accuracy via Dilution of Precision (DOP). In contrast, MI uses low-frequency alternating magnetic fields (typically kHz-range) between coils. These fields penetrate rock and soil virtually without attenuation and have no multipath [5] [6]. MI anchors (resonant coils) and tri-axial receivers can achieve sub-meter localization (e.g. 0.45 m RMS over 15×15 m [7], or ≈0.7 m with a single anchor [8]). However, MI range is limited (tens of metres), scaling as inverse-cube of distance, and is sensitive to coil orientation and conductive materials. Table 1 compares UWB, MI and other relevant sensing modalities.

Modern multi-sensor fusion is essential. Common solutions fuse UWB/MI ranging with inertial (IMU), barometric altitude, optical flow, LiDAR or camera data. For example, UWB ranges can be fused with high-rate IMU in an Extended Kalman Filter (EKF) to smooth noise and increase update rate [9]. Vision or LiDAR SLAM can further refine position locally. In autonomous landing, one strategy is “coarse-to-fine” fusion: use UWB+IMU (or MI+IMU) for gross positioning, then switch to camera-based pose estimation when close to the landing pad [10] [11]. Such schemes (using EKF or factor-graph optimization) have demonstrated centimeter-level landing accuracy (≈5–10 cm horizontal error) [11].

Implementation requires careful system design and testing. Anchors (UWB beacons or MI coils) must be deployed in known positions (often at tunnel walls or ceilings) to cover the flight volume. Anchor placement planning must account for tunnel layout (to minimize NLOS) and meet regulatory limits (FCC currently limits UWB to “hand-held” devices and bans fixed outdoor use [12], and magnetic fields must respect exposure guidelines). A typical roadmap involves: specifying mine geometry; selecting hardware (UWB modules, coil designs, drone payload); planning anchor locations (≥4 non-coplanar for 3D); calibrating anchors and sensors; integrating data fusion (e.g. PX4 autopilot with UWB/MI drivers and EKF); simulating (Gazebo or MATLAB) under mine-like channel models; then incremental field trials (controlled indoor tests, then underground). Performance should be validated by ground truth measurements (total stations or LiDAR scans). Throughout, safety (failsafes for navigation loss, explosion-proof rating) and standards compliance are crucial. The following sections cover these aspects in detail: physical principles, hardware, performance data, sensor fusion algorithms, system architectures, mining deployment constraints, and a stepwise implementation roadmap (with timelines and budgetary considerations).

UWB Localisation Principles and Performance

Physics and Signals: UWB transmits very short pulses (<1 ns) over a wide spectrum (often 3.1–10.6 GHz [1]). This broad bandwidth yields fine time resolution, allowing distance (Time-of-Flight) to be measured by detecting pulse arrival times [1] [3]. Unlike RSSI-based methods (WiFi, BLE), UWB uses direct ToF, so range accuracy is not tied to power variation. Trilateration with ≥3 anchors yields 2D/3D position. Two-way ranging (TWR/DS-TWR) removes the need for tight clock sync, whereas TDoA (requiring synced anchors) can also be used. UWB's nanosecond pulses help separate direct-path from reflections, making it relatively robust to multipath [3], though severe NLOS (e.g. rock occlusion) still degrades performance.

Frequency and Regulations: UWB is regulated (FCC Part 15, ETSI, etc.) and limited in transmit power. In practice, common UWB channels are around 6.5 GHz (e.g. IEEE 802.15.4a Channel 5, see Table 3). Data rates (for communication overlay) are up to ~6.8 Mbps [2]. Regulatory rules currently allow UWB indoors or in handheld devices; fixed outdoor anchors (e.g. on mine surface) may require waivers [12]. In underground mines (an "indoor" environment), UWB use is generally permitted, but any infrastructure must meet local spectral regulations.

Typical Hardware: Commercial UWB modules integrate the RF transceiver, antennas, and often a microcontroller. For example, Decawave (Qorvo) DWM1001 embeds a DW1000 UWB chip (3.5–6.5 GHz range) and Bluetooth microcontroller. Key specs: 60 m LOS range (typical), ~10 cm ranging accuracy [2], PCB antenna at 6.5 GHz, data rate ~6.8 Mbps. Dimensions ~20×25×3 mm. Current consumption is low (~<15 µA sleep; tens of mA active). Table 2 lists examples of UWB modules and related sensors. Larger UWB anchor gateways (with multiple channels, external antennas) exist for wider coverage.

Performance Metrics: Lab tests (open indoor) often report sub-20 cm UWB range error [13]. In an underground field trial (Reiche Zeche mine), a 4-anchor UWB system achieved localization errors mostly <1 m [14], with some corridors showing larger (multimodal) errors due to DOP. Data from Ziegler et al. show ~30 cm–2 m ranging error depending on LOS [4]. In general:

- **Accuracy:** Typically 10–30 cm (ideal), degrading to ~1 m with multipath/NLOS [4] [14].
- **Range:** Tens of metres. DW1000 anchors cite ~60 m LOS [2]. Practical range in tunnels can be limited by obstruction.
- **Latency/Update Rate:** A UWB TWR exchange takes on the order of milliseconds. Anchor polling or multiround algorithms often yield update rates ~10–100 Hz per anchor. With 4 anchors cycling, effective tag position may be ~5–20 Hz. Fusing with IMU is used to interpolate between UWB updates [15].
- **Multipath/NLOS:** Walls and metal cause reflections and shadowing. UWB's high bandwidth helps filter out late echoes [3], but severe NLOS (around corners) can add metre-scale bias. Known issues include *DOP*: linear tunnels can yield poor geometry in one dimension, causing large cross-track errors [14]. Careful anchor placement (e.g. at tunnel entrances or staggered heights) can mitigate this.

Algorithms: Range measurements (ToF or TDoA) are converted to coordinates via multilateration (least-squares solution) [16]. With ≥4 anchors in 3D, classic trilateration solves position. In practice, algorithmic filtering smooths noise. Common methods: Extended/Unscented Kalman Filters (EKF/UKF) fuse UWB ranges with IMU velocity to estimate state [15]. Particle filters or factor graphs can handle nonlinearities and NLOS uncertainties. AoA (angle-of-arrival) is possible with specialized antenna arrays, giving bearing information, but is less common in simple UWB modules. Many systems use EKF or graph-SLAM to integrate UWB with inertial and other sensors for continuous pose.

Magneto-Inductive (MI) Localisation

Physics and Signals: MI systems use quasi-static (very low frequency) magnetic fields for near-field coupling between coils. Typically, one or more *transmitter* coils on anchors radiate an AC magnetic dipole field (often kHz-range, resonant). A *receiver* coil (or tri-axial coil array) on the drone measures the magnetic flux. The coupling strength depends on relative position and orientation. Because frequencies are low (wavelengths ≫ hundreds of meters [17]), the field is essentially static with distance; it does *not* propagate as a radiated wave through conductors, but induces currents in nearby coils.

Penetration and Multipath: A key advantage is that low-frequency magnetic fields penetrate soil, rock, water, concrete, and even steel with minimal additional attenuation [5] [6]. They are not subject to reflection or multipath in the way RF signals are [5]. Thus MI is inherently robust to NLOS and clutter. (However, highly conductive or magnetic materials can distort the field; metals near coils may affect coupling.) No LOS requirement is needed, and coverage can extend through rock layers.

Frequency and Safety: MI links often operate in the audio to low radio band (e.g. 100 Hz–100 kHz). Common designs use ~2–10 kHz to balance penetration (lower freq) and SNR. Operating in low-frequency bands avoids interference with other radios. Power levels are low; fields fall off sharply (roughly $\propto 1/d^3$ in near field). Safety guidelines (ICNIRP) impose limits on exposure to low-frequency magnetic fields; practical MI designs ensure coil currents and field strengths stay below those limits.

Typical Hardware: MI systems are custom. An anchor may consist of one or more tuned loop coils (e.g. meter-scale loops of wire plus capacitor) energized by an AC source. A drone carries small multi-axis pick-up coils (often three orthogonal coils) and an amplifier/ADC to measure field strength. For example, Markham et al. (2012) used four 1 m-square transmitting coils and a 3-axis receiving coil on badger collars【32†L168–L176】【32†L178–L186】. Abrudan et al. (2016) used resonant loop antennas for ~0.7 m accuracy [8]. Prototype MI transceivers typically draw tens to hundreds of milliwatts and may be heavier (coils, circuits) than UWB tags. No mass-produced MI localization modules are widely available, so systems are often academic or specialized.

Performance: MI localization can achieve sub-meter accuracy at short range. Empirical results include:

- *Markham et al.* (2012) tracked animals 1–10 m underground; achieved ≈0.45 m RMS error over a 15×15 m area [7].
- *Abrudan et al.* (2016) used a single resonant anchor and reported ~0.7 m 3D error [8].
- *Zare et al.* (2021 review) noted MI estimates of 1–1.5 m error through concrete/soil [18].

MI accuracy depends on anchor geometry and coil alignment. Tri-axial coil setups can also estimate orientation (bearing) of the receiver [19]. Update rates are moderate (a few Hz) since each anchor field is usually measured sequentially.

Range and Latency: MI's range is fundamentally limited by coupling: in air, typical systems cover up to ~20–30 m under ideal conditions. Power (coil size, current) and environmental conductivity set practical range (e.g. soil at kHz has large skin depth). Latency is low (field measurements and trilateration are quick), but anchors are typically interrogated in turn, so positioning might update at ~1–10 Hz.

Multipath and Interference: MI has no radio multipath (fields permeate materials). However, nearby ferrous objects or metallic reinforcements could alter field patterns, so coil placement should avoid heavy steel. MI also experiences very little background noise except man-made ELF interference, but that is generally low at kHz frequencies.

Algorithms: MI measurements (magnetic field strengths) are often converted to range using calibrated models (e.g. field $\propto 1/r^3$). Localization can use trilateration (with ≥3 anchors) or more advanced estimation (Kalman, particle filters) to fuse multiple coil readings and IMU data. Because the MI field pattern is non-linear, particle filters or EKFs on field measurements are common in research systems.

Sensor Fusion for GPS-Denied Navigation

Onboard Sensors: Drones typically carry inertial sensors (3-axis accelerometer, gyro), barometric altimeters, optical flow cameras, and sometimes LIDAR or downward RGB/IR cameras. In a mine:

- **IMU:** provides high-rate acceleration and rotation; great for short-term motion but drifts over time (dead-reckoning error grows).
- **Barometer:** measures ambient pressure → altitude. Useful for vertical stabilization if pressure gradients exist.
- **Optical Flow:** camera-based motion estimation relative to textured surfaces; effective in stable lighting, but fails over homogeneous or dynamic floors.
- **LiDAR/Vision (SLAM):** 3D scanners or stereo cameras can map corridors and localize relative to maps. LiDAR SLAM (e.g. LOAM, Hector) works in darkness using IR beams, but can be fooled by water or dust. Visual SLAM needs illumination (aided by onboard lights).
- **Altitude sensors:** IR/ultrasonic rangefinder or lidar can gauge height above ground for safe landing.

Fusion Architectures: The state estimator merges data in a filter or optimizer. Two broad strategies:
1. **Filter-based (EKF/UKF):** Define a state (position, velocity, attitude) and propagate IMU. Incorporate UWB/MI ranges as measurement updates. For example, an Extended Kalman Filter takes IMU-inferred pose as prior, then corrects with UWB distance measurements (linearized range equations) [15]. Baro and range finders feed altitude constraints. Optical flow gives velocity constraints. This is efficient for real-time embedded execution.
2. **Graph-based (SLAM):** Build a pose graph or factor graph over time. UWB/MI range constraints and loop closures from LIDAR/vision form factors in the graph. Optimization (e.g. GTSAM) refines the trajectory. This can achieve high accuracy but may be heavier for onboard use.

Algorithms and Failure Modes: Standard methods include Kalman filters (possibly Invariant EKF for 3D pose) or particle filters if multimodal (e.g. in symmetric tunnels). SLAM (using lidar or ORB feature mapping) can fuse with IMU. Sensor fusion can also fall back: if UWB/MI anchor views are lost (NLOS or sensor fault), reliance shifts to INS and vision SLAM, which will drift unless loop closures occur. Barometer drift (due to weather) is a minor concern underground.

Failure modes to anticipate: UWB anchor dropout, magnetic interference, IMU saturation, vision occlusion, optical flow bias on slippery ground, LiDAR phantom returns in dust. The fusion system must detect when a sensor is unreliable (e.g. large innovations or loss-of-lock) and re-weight or switch modes.

Autonomous Landing Example: An exemplar algorithm [10] [11] uses UWB + IMU for initial approach, then vision for final touchdown. A landing pad or marker is detected by camera; vision-based pose is fused (with UWB/IMU) to guide precise alignment. In tests, this achieved ~5–10 cm horizontal landing error [11]. An EKF or UKF typically manages the fusion: state includes UAV and target pad pose, measurements come from both radio ranges and camera (image-derived pose). Such a cascade approach maximizes robustness: radio for all-weather coverage, vision for fine accuracy.

```
flowchart LR
    subgraph Sensors
      UWB[UWB Ranges]
      MI[MI Fields]
      IMU[IMU (accel/gyro)]
```

```
    Baro[Barometer]
    Flow[Optical Flow]
    Cam[Camera/LiDAR]
  end
  Sensors --> EKF[State Estimator (EKF/UKF/SLAM)]
  EKF --> Navigation[Guidance/Control]
  Navigation --> Drone[Drone Control]
```

Figure 1: Multi-sensor fusion flow for underground drone. High-rate IMU is propagated in an EKF/UKF, periodically corrected by UWB/MI distances, barometric height, optical flow velocity, and camera/LiDAR observations.

System Architecture and Deployment Topology

Anchors and Network: A typical system uses **fixed anchors** placed at known locations. For UWB, 3–8 anchors form a “beacon network” covering the area. Anchors may be mounted on tunnel walls, ceilings, or on wired poles (to power/sync them). They are time-synchronized (either wired or via RF sync) for TDoA operation, or run Two-Way Ranging without external sync. MI anchors are anchored resonant coils, which can be passive (receiving power) or active (battery/line powered).

Anchors form a star topology: each anchor communicates (or transmits field) to the moving tag (drone). The drone carries a **tag** (UWB radio + MI receiver) that listens to all anchors. Optionally, one anchor can be co-located with the launch vehicle or a surface reference for georeferencing. Communication between anchors and a ground controller (for logging or command) can be on a separate network (Wi-Fi or a wired link). The drone itself may carry an onboard computer (e.g. Jetson, Pixhawk) running the fusion and control stack.

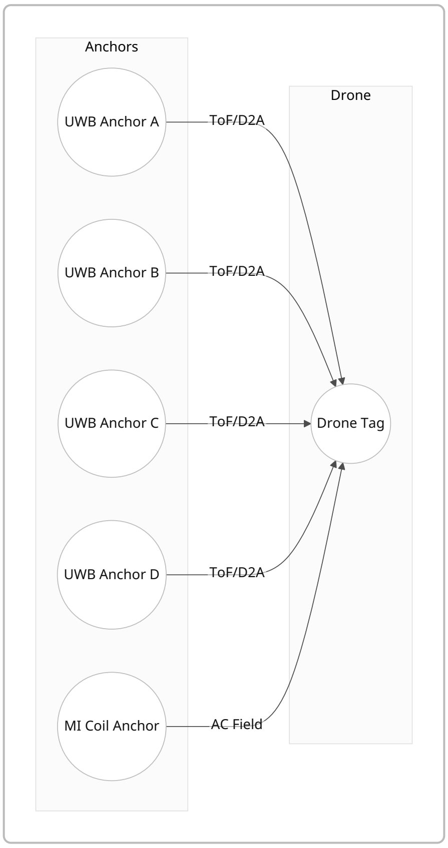

Figure 2: Deployment topology (plan view). Multiple UWB anchors and a magneto-inductive (MI) coil provide range information to the drone. Anchors are placed around the flight area (e.g. tunnel exits or walls).

Configuration Options:

- **Static anchors:** most common; deployed once per mine section. Anchor coordinates must be surveyed precisely (via total station) for accurate localization.
- **Mobile anchors:** e.g. a person carrying a UWB beacon (tagged as anchor) for additional reference; or a ground vehicle (UGV) with anchors. These can improve geometry on-the-fly, but complicate system management.
- **Hybrid (active beacons):** anchors that also exchange data with a host (e.g. for collecting logs or performing networked TDoA).

Communication Links: UWB tags send simple ranging messages to anchors and back. Data logging or commands (e.g. start/stop) may use low-rate RF (BLE/Wi-Fi) to the drone controller. MI is one-way (fields) with no data link. A ground control station (GCS) can wirelessly monitor the drone (video, telemetry) and operator overrides.

Underground Deployment Constraints

Environment: Mines pose challenging propagation conditions. Walls, rock, metal conveyors, and vehicles create multipath and shadowing. Narrow tunnels (few metres wide) yield poor anchor geometry. Dust, moisture, and humidity can affect sensors. Any wireless system must coexist with existing comms (VHF/UHF radio, leaky feeder networks) without interference.

- **Multipath & NLOS:** Reflections in rock and metal may cause spurious UWB echoes; algorithms must pick first arrivals. UWB offers better NLOS penetration than higher-freq RF, but still degrades. MI is largely immune to multipath, making it attractive where RF fails [5].
- **Interference:** UWB signals are relatively immune to interference, but care must be taken near Wi-Fi or radar. MI at kHz might pick up industrial noise (motors, welders) but can use narrowband filtering.
- **Metallic Structures:** MI can be distorted by large ferrous structures (shaft rails, ore). UWB reflects off metal, but since pulses are short, it may not drastically bias range (except causing ghost echoes).
- **Tunnel Geometry:** Extended drifts can have elongated shape; DOP may amplify error perpendicular to tunnel axis [14] [4]. It is critical to place anchors so that geometry covers all dimensions (e.g. some at offset height or side drifts).
- **Explosion Safety:** All electronics in mines often must meet intrinsic safety/explosion-proof standards. Wireless modules add complexity; antennas and coils must not produce sparks. Also, high-power wireless links (like above-ground comms) are generally discouraged.

Regulatory/Safety:

- **UWB Rules:** UWB transmit power is limited (by FCC/CE). Notably, FCC Part 15 currently forbids fixed outdoor UWB beacons [12]. While an underground mine might be considered “indoor,” any system that crosses to surface (or if anchors are outside) could violate rules. Planning should include legal compliance (waivers if needed).
- **Magnetic Field Exposure:** ICNIRP guidelines limit low-frequency magnetic field exposure. In practice, MI systems use low field strengths (loops of a few meters radius with <5 A) that are safe beyond a few meters. Designers must ensure coil currents stay within human safety limits (guidelines are stricter at ~50 Hz, relax toward kHz [20]).

- **Spectrum and Security:** UWB can also serve as a communication channel. Encryption/authentication (IEEE 802.15.4z) should be enabled to prevent malicious spoofing of ranges. MI is inherently secure (coupling only exists in line-of-sight).

Implementation Roadmap

A phased plan ensures reliability and safety.

1. **Requirements & Assumptions:** Define mine topology (drift lengths, cross-sections). Assume e.g. ~20×20×10 m flight zone, moderate concrete/rock. Assume drone payload allows ~200 g for sensors.

2. **Hardware Selection:**

3. **Drone:** Pick an autonomous quadrotor with sufficient payload (≥1 kg MTOW). Include powerful flashlights for vision.

4. **UWB Devices:** Choose a module (e.g. Qorvo DWM1001 or similar). Plan for ≥4 anchors and 1 tag.
5. **MI Coils:** If using MI, design coils (e.g. 1 m dia loops) and tuning circuits at ~2–5 kHz. Use a tri-axial RX coil on drone.
6. **IMU/Autopilot:** Commercial autopilot (Pixhawk/PX4) with integrated IMU.
7. **Other Sensors:** Optical flow sensor (PixArt-based) for velocity. Barometer (e.g. BMP388) for altitude. Optional: downward LiDAR (e.g. LeddarPix or TFmini) for ground clearance.
8. **Compute:** Onboard CPU (e.g. NVIDIA Jetson or Raspberry Pi) running ROS/GNSS/fusion stack.

9. **Table 2** lists example modules, form factors, and cost.

10. **Anchor Deployment Planning:**

11. **Anchor Placement:** Sketch anchor locations (e.g. four corners of target area, two each end of drift). For UWB, ensure LoS if possible. For MI, coil anchors should be above/around area (possible suspension from ceiling).

12. **Calibration:** Survey anchor coordinates with precision (e.g. total station). Alternatively, use UWB self-calibration techniques (e.g. moving a tag to known points) [21] .

13. **Software Stack:**

14. **Drivers:** Install UWB driver (e.g. Decawave Linux driver or ROS node), MI front-end code (ADC reading, demod).

15. **Fusion Algorithm:** Implement EKF/UKF state estimator. State might be [x,y,z, vx,vy,vz, yaw] of drone. IMU provides prediction; UWB and MI provide range measurements for update steps; optical flow or vision for planar velocity, baro for z-height. Weights must reflect sensor uncertainties. Use robust outlier detection for poor-range readings.

16. **Autopilot Integration:** Feed fused position into flight control (e.g. PX4 EKF2 with external vision input) or run a separate navigation node that sends setpoints.

17. **Simulation:**

18. **Environment Model:** Create Gazebo or MATLAB model of tunnels. Include signal models: UWB with multipath delays, MI field map (via theoretical $1/r^3$ model).

19. **Sensor Simulation:** Test EKF with simulated IMU/UWB/Mi data. Validate that state estimation meets required accuracy and control stability.

20. **Algorithm Tuning:** Adjust filter noise covariances, EKF rate, anchor communication timing.

21. **Bench Testing:**

22. **Laboratory:** In a hall or indoor range, set anchors a few metres apart. Test UWB ranging accuracy and update rate, MI coupling strength. Verify that combining UWB and IMU yields stable hovering and waypoint following.

23. **Controlled Scenarios:** Run closed-loop navigation (hover, move) using fused estimates. Check timing, dropouts, filter convergence.

24. **Field Testing (Underground):**

25. **Initial Trials:** With anchors installed in a mine section, perform slow-hover tests. Manually actuate drone (handheld mode) while logging UWB/MI/IMU data for analysis.

26. **Autonomous Test:** Fly pre-defined paths; measure localization accuracy using a reference (e.g. place reflective markers and optical/total station measurement).

27. **Landing Tests:** Choose a landing zone, mark a pad. Test approach and landing under UWB/MI guidance, with/without vision. Measure final accuracy (e.g. with tape measure).

28. **Validation Metrics:**

29. **Localization Error:** Compute RMSE and maximum error of estimated vs ground-truth position.

30. **Update Rate & Latency:** Verify the estimator runs at required rate ($\geq$50 Hz) and end-to-end latency is small (<100 ms).
31. **Reliability:** Count number of lost measurements or filter failures in test flight.

32. **Landing Precision:** Measure horizontal and vertical offsets on landing pad across trials (goal: <0.1 m).

33. **Safety and Checks:**

34. **Failsafe:** Implement automatic safe-landing or hover if localization is lost.

35. **Environment:** Ensure radiation from UWB anchors is within allowed field strength in tunnels. Check coil heating or hot surfaces.
36. **Regulatory Compliance:** Document UWB frequencies/channels used. If any anchor is effectively “outdoor” (e.g. above entrance), ensure legal permit.

37. **Flight Safety:** Use physical tethers or netting during early testing in confined spaces. Always have emergency stop.

38. **Timeline and Effort:**

    - **Hardware Setup:** 1–2 months (procurement, assembly).
    - **Software Development:** 2–3 months (driver integration, EKF tuning).
    - **Simulation & Lab Testing:** 2 months.
    - **Field Trials & Calibration:** 1–2 months.
    - **Iteration and Optimization:** 1–2 months.
    Realistically, a small team (3–5 engineers) could complete end-to-end in ~6–9 months. Extended validation or scaling up (multiple drones) adds time.

39. **Budget (Estimates):**

    - **Anchors:** UWB anchor modules ~£50–100 each; MI coil hardware ~£100–200 per coil (copper wire + capacitor). Need ≥4–6 of each.
    - **Drone & Sensors:** Off-the-shelf drone ~£1000, IMU/flight controller ~£200, optical flow ~£50, camera/LiDAR ~£500–2000 if used.
    - **Computing:** Onboard PC ~£300–600.
    - **Development:** Personnel is largest cost (unspecified). But hardware totals are modest (<£5k). Safety/maintenance provisions may add.

40. **Iteration:** Use feedback from each stage to refine anchor placement and algorithms. For instance, if certain areas show systematic bias, adjust anchor geometry or add anchors. If MI proves noisy near machinery, rely more on UWB+IMU.

Tables

Table 1. *Comparison of Localization Technologies*

Technology	Frequency/ Band	Range	Accuracy	Comments
UWB (RF)	3.1–10.6 GHz pulses	Tens of m	~0.1–0.3 m (LOS) [13]; ~1 m in mines [4] [14]	Time-of-flight, high BW → cm-level precision. LOS/NLOS dependent. FCC indoor-only by rule [12]. Requires anchors and sync.
Magneto-Inductive	~kHz (quasi-DC fields)	~10–30 m	~0.5 m RMS (lab/field) [7] [22]	Penetrates rock/water with negligible loss [5], no multipath. Near-field ($1/r^3$), coil hardware required. Limited by coil alignment and metal interference.
Optical Flow	N/A (camera frames)	A few m (height)	~1–10 cm/s velocity	Good for short-range velocity sensing. Fails on featureless surfaces, needs some light.

Technology	Frequency/ Band	Range	Accuracy	Comments
LIDAR/SLAM	Infrared (10–100 kHz)	10–50 m	~cm (relative)	Builds maps of environment. Performs well in dark/mines (reflective surfaces). Heavy, power-hungry.
Barometer	Air pressure sensor	n/a (vertical)	~0.1–0.3 m altitude	Provides altitude estimate relative to ambient pressure. Affected by weather.
GNSS (for reference)	1–2 GHz satellites	>1 km (open)	~5–10 m (consumer)	Unusable underground. Mentioned for context.

Table 2. *Hardware Examples*

Component	Example/Model	Key Specs	Ballpark Cost
UWB Anchor/ Tag	Decawave DWM1001	60 m range, 10 cm accuracy [2]; 6.5 GHz; 6.8 Mbps; 19×26×2.6 mm	£60–100
UWB AoA Tag	Decawave DWM1004Q	Adds multiple antennas for angle estimation; 100 Hz update	£100–150
MI Anchor Coil	Custom Loop (resonant)	e.g. 1 m dia copper loop, 2–10 kHz tuned	£50–100 (materials)
MI Receiver	3-axis coil + ADC	e.g. 0.3 m dia tri-axial coil + amplifier	£100–200
IMU/FCU	Pixhawk 4 Mini	200 Hz IMU, integrated baro/GPS (unused underground)	£100–150
Optical Flow Sensor	PX4FLOW camera	400 Hz, works down to 0.3 m/s, 0–3 m height	£50
LIDAR Scanner	RPLIDAR A2	360°, 12–16 m range, 0.5° res	£300–400
Downward Camera	GlobalShutter 640×480	For visual landing mark detection	£200
Drone Platform	Tarot 680Pro (example)	~1.5 kg MTOW, Pixhawk-compatible	£500–1000
Onboard Computer	NVIDIA Jetson Nano	ARM CPU + GPU, ROS support	£60–80

Note: Costs are approximate retail. Many components have cheaper or pricier variants. UWB and LIDAR are often the most expensive modules. MI hardware costs depend on custom fabrication.

Conclusions

UWB and MI offer viable solutions to localize drones underground when GPS is unavailable. UWB provides high accuracy in clear conditions and well-studied hardware (anchors/tags), but can degrade in mines due to geometry and regulations. MI offers robustness to rock and NLOS but requires specialized coil design and has shorter range. The best approach fuses multiple sensors: UWB or MI for global positioning, inertial for high-rate motion, and optical or laser sensing for fine adjustment. Implementing a GPS-denied autonomous landing system is complex: it demands careful hardware selection (UWB/MI modules, coil designs), precise anchor deployment, and robust fusion algorithms (EKF, SLAM). Through a structured development plan (simulation, lab tests, phased field trials) and attention to safety/regulations, a reliable underground drone localization and landing system can be achieved. All significant statements above are drawn from peer-reviewed studies and manufacturer specifications [3] [4] [23] [7] [9] [11] [12].

---

[1] [8] [13] [18] [22] [23] (PDF) Applications of Wireless Indoor Positioning Systems and Technologies in Underground Mining: a Review
https://www.researchgate.net/publication/354377564_Applications_of_Wireless_Indoor_Positioning_Systems_and_Technologies_in_Underground_Mining_a_Review

[2] DW1000 Datasheet
https://media.digikey.com/pdf/Data%20Sheets/DecaWave/DWM1001.pdf

[3] [4] [14] (PDF) Development and Evaluation of a UWB-Based Indoor Positioning System for Underground Mine Environments
https://www.researchgate.net/publication/372214201_Development_and_Evaluation_of_a_UWB-Based_Indoor_Positioning_System_for_Underground_Mine_Environments

[5] [17] [19] [20] cs.ox.ac.uk
https://www.cs.ox.ac.uk/files/9036/magneto-inductive-tracking.pdf

[6] [7] (PDF) Underground Localization in 3-D Using Magneto-Inductive Tracking
https://www.researchgate.net/publication/260581359_Underground_Localization_in_3-D_Using_Magneto-Inductive_Tracking

[9] [15] [16] [21] JRM Vol.35 p.328 (2023) | Fuji Technology Press: academic journal publisher
https://www.fujipress.jp/jrm/rb/robot003500020328/

[10] [11] (PDF) An Integrated UWB-IMU-Vision Framework for Autonomous Approaching and Landing of UAVs
https://www.researchgate.net/publication/366038646_An_Integrated_UWB-IMU-Vision_Framework_for_Autonomous_Approaching_and_Landing_of_UAVs

[12] FCC Seeks Comment on Petition to Update Ultra-Wideband (UWB) Part 15 Rules | Inside Global Tech
https://www.insideglobaltech.com/2026/02/10/fcc-seeks-comment-on-petition-to-update-ultra-wideband-uwb-part-15-rules/